

Matching polytopes as exact Pauli uncertainty bodies: all line graphs and a simplicial claw-free frontier

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Abstract

For Pauli observables with a prescribed anticommutation graph, weighted sums of squared expectations need not equal the corresponding weighted independence number. Our main analytic result proves equality for every line graph and every nonnegative weight vector. The proof maps a line-graph Hamiltonian to a Majorana bilinear and shows that squared entries of a skew polar factor satisfy Edmonds' full matching-polytope description. It yields the sharp inequality $\|A\|_*^2/4 \leq \nu(R, w) \sum_e A_e^2/w_e$ and reduces the complete weighted uncertainty problem to maximum-weight matching. This strictly extends the previously sufficient h -perfect route: the infinite family $L(K_{2k+1})$, $k \geq 2$, is not h -perfect but is covered by our theorem. We also give computer-assisted proofs of equality for an explicit family of simplicial claw-free graphs of order $2m + 8$, for every $m \geq 0$ and every nonnegative weight vector. A signed rectangular Gram identity controls one full-support weighted inequality; an exact eight-vertex core and a uniform transportation argument supply the remaining weights. A related three-row identity proves a twelve-vertex case and yields all-weight corollaries for a published XX-strip and one specified closure. We then isolate a twenty-four-vertex composition with a full-support classical facet of bound six. A rational two-copy dual certificate now proves its quantum maximum is exactly six; all 16384 dual coordinates and 8256 symmetric-support inequalities are checked with integer arithmetic. A rational feasible point of the first-moment relaxation has objective 6.50657958. The classical facet has known gear/lifting ancestry, so its polyhedral construction is not claimed new. Exact algebraic and polyhedral acceptance is separated from numerical discovery. The earlier archived bundle passes 127 tests and checks 64 artifact hashes; the new line-graph audit adds 1245 finite root-graph stress cases, 47 direct dense Majorana checks, and five corruption-controlled tests. The results do not establish weighted perfection of all simplicial claw-free graphs or a general quantum strip-composition rule.

Status. Theorem statements below have internal analytic and exact-computation support. External proof review and priority assessment are outstanding. This draft makes no first-discovery or venue-level claim.

1 Introduction

The anticommutation graph records a useful part of the structure of a many-qubit Pauli Hamiltonian without specifying a particular tensor-product representation. For observables $P_i = P_i^\dagger$, $P_i^2 = I$, place an edge between i, j exactly when $P_i P_j = -P_j P_i$; nonedges require commutation. For a graph G and weights $w \geq 0$, consider

$$\beta(G, w) = \sup_{\{P_i\}, \rho} \sum_{i \in V(G)} w_i \langle P_i \rangle_\rho^2, \quad \alpha(G, w) = \max_{S \text{ stable in } G} \sum_{i \in S} w_i. \quad (1)$$

Here the supremum includes Pauli realizations with exactly the stated pairwise relations. All upper bounds in this paper apply to every such realization and every density matrix. A common eigenstate of the commuting operators in a maximum-weight stable set proves $\beta(G, w) \geq \alpha(G, w)$.

The weighted beta number, the equality termed $\hbar$ -perfection, and facet-based reductions are established concepts [7]. The free-fermion solution for simplicial claw-free (SCF) frustration graphs provides our spectral input [6]. Neither that solution nor the unweighted bound by itself resolves the weighted problem: a stable-set polytope can have non-rank facets. The contribution examined here is a specific route from coherent, sign-dependent transfer coefficients to all-weight bounds on explicitly defined graphs.

The paper has three different kinds of outcome. Section 3 gives an analytic arbitrary-size theorem. Sections 4–6 give proved, restricted results. Section 7 gives an exact fixed-weight quantum certificate for a composed graph, alongside its classical ancestry and earlier failed numerical attempts. The unresolved question is now a general quantum composition rule, not the truth of that single inequality. Numerical nonviolations are kept separate from upper-bound proofs.

The construction uses known classical ideas: stable-set facets, matching, transportation, and the published XX-strip. We do not claim new graph decompositions or a new quantum-simulation algorithm. In particular, solving a constituent atom is not a theorem about arbitrary strip compositions. Earlier QAOA/MPS performance and symmetry investigations are outside this manuscript and supply no evidence for its claims.

2 Preliminaries and rank-to-weight reduction

A stable set is a set of pairwise nonadjacent vertices, including the empty set. Its incidence vector is $\mathbf{1}_S$, and $\text{STAB}(G) = \text{conv}\{\mathbf{1}_S : S \text{ stable}\}$. This polytope is full-dimensional and downward closed in the nonnegative orthant. Thus proving that every squared profile $x_i = \langle P_i \rangle_\rho^2$ satisfies its complete nonnegative-weight description proves all-weight equality in (1). Attainability is then established separately by the commuting stable-set argument.

A graph is claw-free if it has no induced $K_{1,3}$. A nonempty clique K is simplicial if $N(v) \setminus K$ is a clique for every $v \in K$. We use SCF componentwise: every connected component is claw-free and admits such a clique. Induced subgraphs remain SCF componentwise; this is the inherited structural class used by the free-fermion framework [6]. The finite support checks in our certificates also verify the necessary SCF witnesses directly.

For fixed observables the variational identity

$$\max_{\rho} \sum_i w_i \langle P_i \rangle_{\rho}^2 = \max_{\|a\|_2=1} \left\| \sum_i a_i \sqrt{w_i} P_i \right\|^2 \quad (2)$$

follows by duality of the Euclidean norm and then maximization over states. Equivalently, for positive weights the Hamiltonian amplitudes $b_i = a_i \sqrt{w_i}$ obey $\sum_i b_i^2 / w_i = 1$. Zero weights are omitted or obtained by continuity.

Lemma 2.1 (SCF rank bound). *For SCF G and every induced support U , $\sum_{i \in U} \langle P_i \rangle_{\rho}^2 \leq \alpha(G[U])$.*

Proof. It suffices to work in the universal Pauli algebra and then restrict to physical representations. Write $A = \sum_i b_i P_i$ and let Q_k be the sum of $\prod_{i \in S} b_i P_i$ over stable k -sets. Within a stable set the factors commute. For $T(u) = \sum_k (-u)^k Q_k$,

$$A^2 = \left(\sum_i b_i^2 \right) I + 2Q_2, \quad T(u)T(-u) = I - u^2 \left(\sum_i b_i^2 \right) I + O(u^4).$$

The SCF sector theorem [6] gives at most $\alpha(G)$ nonnegative single-particle energies ν_j , with $T(u)T(-u) = \prod_j (1 - u^2 \nu_j^2)$ in a sector and Hamiltonian norm at most $\sum_j \nu_j$. The displayed coefficient implies $\sum_j \nu_j^2 = \sum_i b_i^2$. Cauchy–Schwarz and (2) give the rank bound. Apply it to each component of the induced support. Degenerate amplitudes follow by continuity. $\square$

Lemma 2.2 (All weights at independence number two). *For any graph with $\alpha(G) = 2$,*

$$\sup_{w \geq 0, w \neq 0} \frac{\beta(G, w)}{\alpha(G, w)} = \frac{\beta(G, \mathbf{1})}{2}.$$

Consequently every SCF graph with independence number at most two is $\hbar$ -perfect.

Proof. Normalize $\alpha(G, w) \leq 1$. The weight polytope is $0 \leq w_i \leq 1$, $w_i + w_j \leq 1$ for every nonedge. Its vertices are half-integral: a bipartite component of tight equations among fractional coordinates admits a nonzero alternating perturbation, while an odd cycle forces every coordinate in its component to be $1/2$. At an extreme weight, let H have weight one and L weight one-half. Then H is a clique completely joined to L . Cross anticommutation gives $(A_H + A_L)^2 = A_H^2 + A_L^2$, and hence the beta bound on a complete join is the maximum of the bounds on its parts. With $b = \beta(G, \mathbf{1}) \geq 2$, the extreme-weight bound is at most $\max(1, b/2) = b/2$. Convexity in w extends it to the weight polytope; uniform weights attain the ratio. Lemma 2.1 gives $b = 2$ in the SCF case. Cliques are immediate. The half-integral and join tools are standard; no new polyhedral principle is asserted. $\square$

3 All line graphs are $\hbar$ -perfect

Let $R = (V, E)$ be a finite simple graph and $L(R)$ its line graph: the vertices of $L(R)$ are the edges of R , with adjacency when two root edges share an endpoint. Write $\nu(R, w)$ for the maximum weight of a matching. Then $\alpha(L(R), w) = \nu(R, w)$.

Lemma 3.1 (Skew trace norm controlled by matching). *Let $w_e > 0$ and let $A = -A^\top$ be a real matrix supported on $E(R)$, with upper-triangular entry A_e on edge e . Then*

$$\left(\frac{\|A\|_*}{2} \right)^2 \leq \nu(R, w) \sum_{e \in E} \frac{A_e^2}{w_e}. \quad (3)$$

The constant is sharp for every R, w .

Proof. By nuclear/operator-norm duality, choose a real Q with $\|Q\|_{\text{op}} \leq 1$ and $\langle A, Q \rangle_F = \|A\|_*$. Replacing Q by $(Q - Q^\top)/2$ preserves the pairing with skew A and does not increase the operator norm. Hence take Q skew and put $x_e = Q_e^2$.

For each vertex v , the incident sum $\sum_{e \in \delta(v)} x_e$ is at most the squared norm of row v of Q , hence at most one. If $S \subseteq V$ has odd cardinality, the odd-order skew principal matrix $Q[S]$ has rank at most $|S| - 1$. All its singular values are at most one, so

$$2 \sum_{e \in E(S)} x_e \leq \|Q[S]\|_F^2 \leq |S| - 1.$$

Thus x satisfies the nonnegativity, degree and odd-set inequalities in Edmonds' matching-polytope theorem [1]. It follows that $\sum_e w_e x_e \leq \nu(R, w)$. Since skew symmetry gives $\|A\|_*/2 = \sum_e A_e Q_e$, weighted Cauchy–Schwarz proves (3). For sharpness, support A on a maximum-weight matching and take A_e proportional to w_e . $\square$

Theorem 3.2 (Line-graph $\hbar$ -perfection). *For every finite simple graph R and every nonnegative edge weight w ,*

$$\beta(L(R), w) = \alpha(L(R), w) = \nu(R, w).$$

Consequently every finite line graph is $\hbar$ -perfect.

Proof. Zero-weight edges may be deleted. Assign a Majorana operator γ_v to each root vertex and represent $e = uv$ by the Hermitian involution $P_e = i\gamma_u\gamma_v$. This is a Pauli realization after a Jordan–Wigner transform, and its frustration graph is $L(R)$ [2]. Representation invariance of the weighted beta number [7] permits this choice.

For real coefficients c_e , let A be the skew matrix with entries $A_e = c_e$. Orthogonal skew normal form shows

$$\left\| \sum_e c_e P_e \right\| = \frac{\|A\|_*}{2}.$$

Combining the variational identity (2) with Lemma 3.1, under $\sum_e c_e^2/w_e = 1$, proves the upper bound $\beta \leq \nu$. A maximum-weight matching is a stable set of $L(R)$; its Pauli involutions commute and a joint eigenstate attains ν . Hence equality. $\square$

There is an independent polyhedral derivation. Every line graph is SCF: it is claw-free, and a root star is a simplicial clique. Apply Lemma 2.1 to the line-graph support corresponding to a root star to obtain the degree inequality. Apply it to the support $E(S)$ induced by an odd root-vertex set S to obtain

$$\sum_{e \in E(S)} \langle P_e \rangle_\rho^2 \leq \nu(R[S]) \leq (|S| - 1)/2.$$

Together with nonnegativity, these are Edmonds’ complete matching-polytope inequalities. This gives a second proof of Theorem 3.2. The polar argument remains useful because it yields the sharp coefficient-dependent matrix inequality (3) directly.

Corollary 3.3 (Strictly beyond h -perfect graphs). *For every $k \geq 2$, $L(K_{2k+1})$ is $\hbar$ -perfect but not h -perfect.*

Proof. The first statement is Theorem 3.2. For the second, assign $x_e = 1/(2k)$ to every edge of K_{2k+1} . Maximal cliques of its line graph are root stars and triangles; their values are respectively one and $3/(2k) \leq 1$. Every odd cycle of length $\ell \geq 5$ has value $\ell/(2k) \leq (\ell - 1)/2$. Thus x satisfies nonnegativity, clique and odd-hole inequalities. Its total value is $(2k + 1)/2$, while every matching has size at most k . It therefore violates the full odd-set matching inequality by $1/2$, excluding h -perfection. $\square$

The theorem supplies an exact polynomial-time uncertainty oracle for all line-graph Pauli families: compute a maximum-weight matching rather than simulate a state or solve a noncommutative SDP. It also identifies the convex mechanism missing from clique and odd-hole bounds: odd principal skew matrices lose one rank, producing every blossom constraint at once. The finite C038 campaign verifies the polar-factor inequalities on all 1245 nonempty NetworkX atlas roots, compares the trace-norm formula with dense Jordan–Wigner matrices in 47 small-root cases, and checks the strictness witnesses $L(K_5), L(K_7), L(K_9)$ exactly. The largest direct norm discrepancy is 7.11×10^{-15} . These checks guard constants and indexing; the arbitrary-size conclusion is analytic.

The closest published structures do not by themselves state this result. Chapman and Flammia identify line-graph Hamiltonians with quadratic Majorana models [2]. Zurel, Cohen, and Raussendorf use $L(K_{2n+1})$ to construct vertices of a Pauli simulation polytope [3]; their

objective is magic-state simulation, not the weighted beta body. McNulty studies measurement-incompatibility robustness on line graphs and obtains skew-energy bounds [4], a different optimization invariant. Classical work relates skew energy and matching number for unweighted oriented graphs [5]; consequently the matrix inequality should not be marketed as the first connection between these objects. Our priority-sensitive claim is only the exact all-weight Pauli beta/matching identity and its blossom-polytope proof. The repository contains the search protocol and unresolved priority risks; external expert review remains necessary.

4 A signed Gram bound beyond a single graph

4.1 The rectangular family

For an integer $m \geq 0$, define light vertices by occupied cells in three rows: $A_j = (0, j)$, $B_j = (1, j)$ for $0 \leq j \leq m$, together with $Z = (2, 0)$, $U = (0, m+1)$, $V = (1, m+2)$. Two light vertices are adjacent exactly when they share a row or a column. Add a clique H_0, H_1, H_c . Its light nonneighbors are row zero, row one, and column zero, respectively. Call the resulting graph G_m ; its order is $2m+8$.

Every independent triple contains Z and one noncentral vertex from each other row, in distinct columns. A stable set containing a heavy vertex has at most one light vertex. Thus $\alpha(G_m) = 3$. The clique $\{A_0, B_0, Z\}$ is simplicial. Claw-freeness can be checked uniformly: a possible claw involves at most four noncentral column indices, so it embeds into G_4 by renaming those columns. The exact four-vertex check on G_4 is exhaustive. This is a finite-support proof, not extrapolation from graph sizes.

Theorem 4.1 (Uniform full-support bound). *For every $m \geq 0$, every Pauli realization of G_m , and every state,*

$$\sum_{i \text{ light}} \langle P_i \rangle_\rho^2 + 2(\langle P_{H_0} \rangle_\rho^2 + \langle P_{H_1} \rangle_\rho^2 + \langle P_{H_c} \rangle_\rho^2) \leq 3. \quad (4)$$

Equality is attainable. The associated classical inequality is a facet of $\text{STAB}(G_m)$.

4.2 Exact transfer coefficients

Use real amplitudes a_j, b_j, c, p, q on A_j, B_j, Z, U, V , and r_0, r_1, r_c on the heavy vertices. Define

$$K_j = -P_{H_0} P_{H_1} P_{A_j} P_{B_j}.$$

Each K_j is a central Hermitian involution in the full generator algebra. Commuting a generator through its four factors gives an even number of sign changes. The square and adjoint are also verified exactly. In an actual common eigenspace write $K_j = \epsilon_j I$ and form

$$B = \begin{pmatrix} a_0 & a_1 & \cdots & a_m & p & 0 \\ \epsilon_0 b_0 & \epsilon_1 b_1 & \cdots & \epsilon_m b_m & 0 & q \\ c & 0 & \cdots & 0 & 0 & 0 \end{pmatrix}, \quad M = B B^\top. \quad (5)$$

Set $L = \text{tr } M$ and $H = r_0^2 + r_1^2 + r_c^2$. Sector signs are retained coherently in this single real matrix. They are not optimized independently, and we do not assume every abstract sign assignment is a physical sector.

For $T(u) = I - uQ_1 + u^2Q_2 - u^3Q_3$, exact expansion in the universal algebra gives

$$T(u)T(-u) = I - e_1 u^2 + e_2 u^4 - e_3 u^6, \quad (6)$$

$$e_1 = L + H,$$

$$e_2 = e_2(M) + (r_0, r_1) M_{\{0,1\},\{0,1\}} (r_0, r_1)^\top + r_c^2 \|B_{:,0}\|^2, \quad (7)$$

$$e_3 = \det M.$$

Here $e_2(M)$ is the sum of its principal two-by-two minors. The exact acceptance algorithm reorders adjacent generators and compares integer amplitude polynomials, including every zero odd coefficient. The discovery algorithm instead uses inversion parity.

These finite checks prove the identities for arbitrary m . A coefficient of degree at most four involves at most four noncentral columns. At degrees five and six, a term comes from a pair and a triple, or two triples; each triple contains Z . At most four other vertices remain. On the matrix side the degree-six determinant has a factor c^2 and the same bound. Relabel the involved columns into G_4 , set unused amplitudes to zero, and transport the word-ordering signs. Thus any failed coefficient at any size would already fail the exact G_4 check.

4.3 The scalar envelope and the spectral branch

Lemma 4.2 (Three-mode Gram envelope). *Suppose an SCF Hamiltonian has at most three sector modes and, in every physical sector, a real positive semidefinite 3×3 matrix M with trace L such that*

$$e_1 = L + H, \quad e_2 \leq e_2(M) + H\lambda_{\max}(M), \quad e_3 = \det M.$$

If $2L + H = 1$ and $L, H \geq 0$, then the Hamiltonian norm squared is at most $3/2$.

Proof. Let $x, y, z \geq 0$ be the eigenvalues of M , with y largest. Then $x + y + z = L \leq 1/2$ and

$$xy + xz + yz + (1 - 2L)y + 2\sqrt{\frac{3}{2}xyz} \leq \left(\frac{1}{4} + \frac{L}{2}\right)^2. \quad (8)$$

For fixed L, y , the left side increases with xz . Replace $x = z = (L - y)/2$ and write $y = s^2/6$. The difference between the right side and the resulting left side is exactly

$$\frac{(s - 1)^2(12L + s^2 + 6s + 3)}{48} \geq 0.$$

For the nonnegative sector energies put $S = \nu_1 + \nu_2 + \nu_3$, padding zero modes, and define

$$f(t) = \left(\frac{t^2 - e_1}{2}\right)^2 - e_2 - 2t\sqrt{e_3}.$$

The symmetric identities in ν_j^2 imply $f(S) = 0$. Equation (8) yields $f(\sqrt{3/2}) \geq 0$. For $t \geq \sqrt{3/2}$,

$$f'(t) = t(t^2 - e_1) - 2\sqrt{e_3} > 0,$$

because $e_1 = 1 - L \leq 1$ and $e_3 \leq (L/3)^3 \leq 1/216$. Consequently $S \leq \sqrt{3/2}$, on the required spectral branch. $\square$

Proof of Theorem 4.1. The two heavy contributions in (7) are Rayleigh forms, bounded together by $H\lambda_{\max}(M)$; use $\lambda_{\max}(B^\top B) = \lambda_{\max}(M)$ for the column norm. For half the weights in (4), the variational normalization is $2L + H = 1$. Lemma 4.2 proves the upper bound $3/2$; rescaling proves (4). The commuting pair $\{H_0, U\}$ has total weight three and gives attainment.

For the facet assertion, an affine functional constant on all tight stable sets must have equal coefficients within each row, by the pairs $\{H_0, A_j\}, \{H_0, U\}$ and $\{H_1, B_j\}, \{H_1, V\}$. The pairs $\{H_c, A_0\}, \{H_c, B_0\}, \{H_c, Z\}$ equate both row coefficients and that of Z , say to a . The triple $\{U, V, Z\}$ sets the right side to $3a$ and every heavy coefficient to $2a$. Thus the tight affine hull has codimension one. $\square$

5 All weights on the unbounded family

Theorem 5.1. *For every $m \geq 0$ and every $w \in \mathbb{R}_+^{2m+8}$, $\beta(G_m, w) = \alpha(G_m, w)$.*

The argument below combines a fixed exact polyhedral lemma with a uniform classical refinement. Quantum validity is checked after the complete classical description is established; no quantum gluing rule is assumed.

For a candidate marginal $x \geq 0$, abbreviate

$$\begin{aligned} a &= x_{A_0}, & b &= x_{B_0}, & c &= x_Z, & p &= x_{H_0}, & q &= x_{H_1}, & r &= x_{H_c}, \\ A &= x_U + \sum_{j=1}^m x_{A_j}, & B &= x_V + \sum_{j=1}^m x_{B_j}, & C &= a + b + c + p + q + r. \end{aligned}$$

The use of A, B here denotes scalar marginals, not operators or the matrix in (5). Write $y = (a, b, c, A, B, p, q, r)$.

Proposition 5.2 (Uniform classical description). *The vector x belongs to $\text{STAB}(G_m)$ if and only if $x \geq 0$, $y \in \text{STAB}(G_0)$, and for $1 \leq j \leq m$,*

$$x_{A_j} + x_{B_j} + r \leq 1, \quad x_{A_j} + x_{B_j} + C \leq 2. \quad (9)$$

The positive inequalities describing $\text{STAB}(G_0)$ are in Table 1; coordinate nonnegativity is also required.

Type	Left sides	Bound
Clique	$a + b + c$; $a + c + q$; $a + A + q$; $b + c + p$; $b + B + p$; $c + p + q$; $A + q + r$; $B + p + r$; $p + q + r$	1
Rank	$C + A$; $C + B$; $C - c + A + B$	2
Weighted	$a + c + A + p + 2q + r$; $b + c + B + 2p + q + r$	2
Full	$a + b + c + A + B + 2(p + q + r)$	3

Table 1: The fifteen positive core inequalities. The eight coordinate inequalities complete the 23-facet description.

Lemma 5.3 (Exact core endpoint). *Among stable-set distributions of G_0 with marginal y , the minimum probability z of selecting both private vertices is*

$$z_{\min} = \max(0, A + B + r - 1, A + B + C - 2). \quad (10)$$

Proof. Lift every one of the 22 stable incidence vectors by the indicator that both private vertices are selected. The complete nine-dimensional lifted hull has 24 facets. Its only lower- z facets are the three in (10); the others are upper- z or z -independent facets. This finite assertion is certified by full-cube rational clipping and facet-rank checks, independently of rational double-description discovery. The exact artifact and acceptance command are given in Appendix B.

Any feasible decomposition of y supplies a value z^* . Lower it to $z_{\min}$. All lower constraints hold, upper constraints remain valid, and the point stays in the cube since $0 \leq z_{\min} \leq z^* \leq 1$. Completeness of the lifted hull proves feasibility and hence the formula. $\square$

Proof of Proposition 5.2. Collapse the first and second noncentral row cliques to their private representatives. Stable sets project to stable sets of G_0 , and (9) is valid by the indicated clique and rank-two supports. This proves necessity.

For sufficiency choose the core distribution of Lemma 5.3 and let $T = A + B - z_{\min} = \min(A + B, 1 - r, 2 - C)$. Remove the absent/absent mass $1 - T$. On the remaining mass construct a transportation table with row supplies $(x_{A_1}, \dots, x_{A_m}, x_U, T - A)$ and column demands $(x_{B_1}, \dots, x_{B_m}, x_V, T - B)$. The last labels mean absence. Forbid (A_j, B_j) and the absent/absent cell, allowing every other cell. Each row forbids at most one column and these forbidden columns are distinct. Every set of at least two rows therefore sees every column. The weighted Hall conditions reduce to $x_{A_j} + x_{B_j} \leq T$ and $(T - A) + (T - B) \leq T$. The first follows from (9) and nonnegativity, the second from $z_{\min} \geq 0$. Ordinary max-flow/min-cut thus provides a table.

Its joint-present mass is $z_{\min}$ and its single-present masses are $A - z_{\min}, B - z_{\min}$. Condition within each of the four core presence patterns to choose actual row labels according to the table. Every label in a row has identical fixed-core neighbors, and forbidden matching pairs never occur. All outcomes are stable and have marginal x . Zero-mass patterns require no conditioning; $T = 0$ is immediate. $\square$

Proof of Theorem 5.1. Every positive core row except the full row omits at least one of c, A, B . After expansion its support has independence number at most two, because every triple of G_m uses Z and both noncentral rows. The stable bound is unchanged by expansion: projection gives one direction and choosing the private labels gives the other. Thus Lemma 2.2 proves these expanded rows quantum-mechanically. It also proves both per-column rows in (9); their supports have independence number at most two. Theorem 4.1 proves the full row. Every squared profile therefore satisfies the complete system of Proposition 5.2, and lies in $\text{STAB}(G_m)$. This proves every nonnegative weight, with attainment as before. $\square$

6 Three-row and published XX corollaries

6.1 A fixed three-row Gram identity

Let J have all nine light cells (r, c) , $0 \leq r, c \leq 2$, with the same row/column adjacency and heavy selectors as G_m . Order the lights row-major, followed by H_0, H_1, H_c ; equivalently its graph6 string is $\text{K}\{\text{S}\{\text{aSfF}\sim\text{Fl}\text{n}\}$. Write the light operators as A_j, B_j, C_j . Besides $K_j = -P_{H_0}P_{H_1}P_{A_j}P_{B_j}$ define

$$\Lambda_0 = I, \quad \Lambda_j = -P_{A_0}P_{C_0}P_{A_j}P_{C_j} \quad (j = 1, 2).$$

All five nontrivial operators are central Hermitian involutions in the full twelve-generator algebra. In each physical sector construct the three rows $(a_j), (\epsilon_j b_j), (\lambda_j c_j)$ of a real 3×3 matrix B , where $\lambda_0 = 1$. Equations (6)–(7) hold unchanged with $M = BB^\top$. The independent acceptance expands all amplitude polynomials, computes e_2 by squared two-by-two minors of B and $e_3 = \det(B)^2$, and checks all odd coefficients. The three even coefficients have 12, 39, and 21 nonzero monomials in the archived canonical encoding.

Corollary 6.1. *The full-support inequality with weights one on the nine lights and two on the three heavies has exact bound three on J . Furthermore, $\beta(J, w) = \alpha(J, w)$ for every $w \geq 0$.*

Proof. The identical Rayleigh estimate and Lemma 4.2 prove the full row. A commuting light/heavy pair attains three. The exact complete stable-set hull has 46 vertices and 36 facets: 12 coordinate facets, 20 rank facets, three weighted facets on SCF supports with independence number at most two, and the full row. Lemmas 2.1 and 2.2 cover the other positive facets. Completeness therefore implies the all-weight claim. The finite algebra and hull obligations are independently checked, not inferred from floating-point optimization. $\square$

6.2 XX-strip and one specified closure

Use the published thirteen-vertex XX construction of Chudnovsky–Seymour [9]; the exact adjacency is reproduced in Appendix A. Call it X . Let X^+ add vertices 14, 15 and only the edges 7–14–15–8, with original one-based labels.

Corollary 6.2. *Both X and X^+ are $\hbar$ -perfect. The result includes every induced subgraph, in particular the eight published variants obtained by deleting any subset of 11, 12, 13 from X .*

Proof. The complete exact hull counts and routes are shown in Table 2. In each graph the sole additional weighted row is

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + 2x_9 + 2x_{10} + x_{11} + x_{12} + x_{13} \leq 3. \quad (11)$$

Its support embeds into J with one heavy vertex omitted. The map from zero-based J to one-based X is

J	0	1	2	3	4	5	6	7	8	10	11
X	1	11	6	13	4	5	2	3	12	9	10.

Both edges and nonedges are independently checked. Corollary 6.1 therefore proves (11); the rank and independence-two lemmas cover all other positive rows. Complete facet coverage proves the claim. Heredity gives the deletion variants. $\square$

Graph	n	α	Stable sets	Facets	Coord.	Rank	Other
J	12	3	46	36	12	20	3 + 1
X	13	4	85	33	13	17	2 + 1
X^+	15	4	203	44	15	26	2 + 1

Table 2: Exact complete hulls. “Other” separates independence-two weighted supports from the single Gram-supported row.

The stable quadruple $\{3, 6, 7, 8\}$ in X excludes an induced embedding into any G_m or J , since those have independence number three. Nevertheless these are corollaries, not independent operator mechanisms. No arbitrary path extension, fuzzy modification, or strip-composition closure is asserted.

7 A fixed composition theorem and its frontier

Take two copies of X with vertices 7, 8 removed. In each copy let $A = \{1, 2, 13\}$ and $B = \{4, 5, 13\}$ in original labels. Join the two B -cliques completely. Add vertices u, v with edge uv , join u to the left A -clique and v to the right A -clique, and add no other edges. Call this twenty-four-vertex graph Y .

For its zero-based ordering, list the eleven retained original labels 1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13 in the left copy, then the right copy, then $u = 22, v = 23$. Freeze weights $w_i = 1 + \mathbf{1}_{i \in \{6, 7\}}$. The exact graph6 and Pauli realization are archived with this ordering.

Proposition 7.1 (Exact classical target). *The graph Y is SCF, has $\alpha(Y) = 6$, and has 2167 stable sets. The inequality*

$$\sum_{i=0}^{23} x_i + x_6 + x_7 \leq 6 \quad (12)$$

is a full-support facet of $\text{STAB}(Y)$, with 88 tight stable sets of homogeneous rank 24.

Proof. Exact enumeration by recursive graph bitsets agrees with a separate product construction from all stable sets of the two eleven-vertex cores and the four connector choices. Both enumerate every stable set, compute the same maximum six, and agree on all tight sets. Independent Fraction elimination verifies the homogeneous rank. SCF witnesses are checked from the graph, including the simplicial clique $\{u, v\}$. The data and checkers constitute the finite certificate. $\square$

Because the tight affine span has codimension one, (12) cannot be a nonnegative linear combination of valid inequalities on proper vertex supports: each positively weighted summand would have to be tight on all 88 points and hence define the same full-support hyperplane. This excludes that particular proof route, not nonlinear reductions or existing structural theorems.

Theorem 7.2 (Exact composed quantum facet, C031). *For the fixed graph Y and weights above, $\beta(Y, w) = 6$. Every finite-dimensional realization with the prescribed exact pairwise relations and every state satisfy*

$$\sum_{i=0}^{23} \langle P_i \rangle_\rho^2 + \langle P_6 \rangle_\rho^2 + \langle P_7 \rangle_\rho^2 \leq 6. \quad (13)$$

The binary adjacency matrix has rank 14 over $\mathbb{F}_2$, and a seven-qubit Pauli realization is explicitly checked for every pair. This is a local simulation, not a hardware run.

Proof. Put $d = 128$ and label Bell coordinates by $v = (x, z) \in \mathbb{F}_2^{14}$. Write $\tau(v) = (-1)^{x \cdot z}$ and let $[v, p]$ be the binary symplectic form. For the archived Pauli labels p_i , define

$$c_v = \sum_i w_i \tau(p_i) (-1)^{[v, p_i]}, \quad A_{vu} = \frac{\tau(v + u)}{128}.$$

The matrix A is real symmetric and implements partial transposition on Bell-diagonal states; its tensor-product and Walsh-transform implementations are checked independently with integers.

Convexity permits maximizing the weighted squared expectations over pure states. Twirling two identical copies by identical Pauli operators preserves the objective and symmetric support, and produces a Bell-diagonal probability vector λ satisfying

$$\lambda \geq 0, \quad \sum_v \lambda_v = 1, \quad \lambda_v = 0 \text{ } (\tau(v) = -1), \quad A\lambda \geq 0.$$

The last condition holds because the twirled state is separable. The two-copy symmetric-support PPT relaxation is established prior art [8, Appendix B.2, Eqs. (42)–(45)], not a new method here.

The archived rational vector u has a common denominator of 332 bits. Exact acceptance checks $u_v \geq 0$ for all 16384 coordinates and $(c + Au)_v \leq 6$ for all 8256 coordinates with $\tau(v) = 1$. Thus, without a solver tolerance or rounding argument,

$$c^\top \lambda \leq (c + Au)^\top \lambda \leq 6.$$

The verifier regenerates c from the archived operators, checks the source hash, and evaluates Au by two agreeing integer algorithms. The common eigenstate of a maximum-weight stable set supplies the matching lower bound six. Finally, weighted representation invariance [8, Theorems 10 and 12] transfers this equality from the archived realization to all realizations in the stated scope. $\square$

The certificate is a computer-assisted proof for one graph and one weight vector, not an all-weight or general SCF theorem. Discovery used nearly active constraints only to select a square correction system; those choices are not trusted by acceptance. Dixon lifting reconstructed the rational correction at 40 lifts (640 modulus bits). Full-coordinate checks, including constraints outside that system, prove the bound. The portable certificate and standalone checker are listed in Appendix B.

7.1 A bounded physical attack

The registered attack fixes the amplitudes of its initial Hamiltonian proposal from one first-moment relaxation and tests 1024 sign classes modulo Pauli-conjugation gauge. The count is 2^{24-14} ; it is a finite set of initializations, not an enumeration of continuous quantum states. At each step it selects an extremal absolute-eigenvalue eigenvector of $H(a) = \sum_i \sqrt{w_i} a_i P_i$ and updates a_i proportional to $\sqrt{w_i} \langle P_i \rangle_\rho$. The cap is 64 iterations per start and the stopping tolerance is 10^{-10} . The threshold for a numerical violation candidate is $6 + 10^{-7}$.

All 1024 starts finished their allocated work. Only 144 met the convergence test; 880 reached the iteration cap. The best saved state, reevaluated with independent bitwise Pauli action, gives 6.000000000000059, consistent with six to floating-point precision. The positive control on the archived eight-vertex G_8 of Xu et al. [7, Fig. 11(a)] gives $3.0448154998549777 > 3$; it is labeled `positive_G8` in the code. The double-weight control, with weights two also at 17, 18, gives 7.000000000000041, consistent with seven. Its upper bound seven already follows by summing two atom rows and the connector-edge bound.

One dense eigensolver failed after a saved 64-start prefix. A second registered batch resumed without skipped classes; twelve resumed-start calls used a checked SciPy fallback. The resumed batch took 214.594 seconds. The saved best and control states were reevaluated independently, but all optimizer trajectories were not independently replayed. These qualifications preclude a global-optimality claim.

7.2 Exact weakness of the first-moment baseline

Define the relaxation explicitly, with indices $0, \dots, 24$:

$$\begin{aligned} \vartheta(Y, w) = \max_M \quad & \sum_{i=0}^{23} w_i M_{i+1, i+1} \\ \text{subject to} \quad & M \succeq 0, \quad M_{00} = 1, \\ & M_{0, i+1} = M_{i+1, i+1} \quad (0 \leq i < 24), \\ & M_{i+1, j+1} = 0 \quad \text{for every edge } ij. \end{aligned} \tag{14}$$

Every physical squared profile is feasible in this relaxation. For a pure state, use real Gram vectors $v_0 = \psi$ and $v_{i+1} = \langle P_i \rangle_\rho P_i \psi$: their real inner products obey the diagonal equations, and the edge products have zero real part. Purification extends this construction to mixed states.

Proposition 7.3 (Rational relaxation witness). *There exists a rational feasible matrix in (14) with*

$$\sum_i w_i M_{i+1, i+1} = \frac{325328979}{50000000} = 6.50657958.$$

In particular, this relaxation cannot certify the bound six.

Proof. The archived matrix has common denominator 10^8 . The acceptance algorithm checks symmetry and every affine equation exactly, and computes a Fraction $LDL^\top$ factorization with 25 positive diagonal pivots and exact reconstruction. The objective is the stated rational number. This finite positive-definiteness certificate proves feasibility without relying on the numerical solver's status. $\square$

The witness proves a relaxation gap of $25328979/50000000$ above the classical bound. It is *not* a quantum state and does not refute (13). Conversely, the physical attack does not give an upper bound that closes this gap; Theorem 7.2 closes it by a separate exact certificate.

The exact degree histogram of Y is $\{4 : 2, 6 : 12, 7 : 4, 8 : 6\}$. In ordinary basic gear composition the two unattached gear hubs have degree five [10]. Thus Y is not a direct output

of that precise construction. General geared, fuzzy, lifting and strip arguments are not excluded by this degree test.

7.3 Classical ancestry and the remaining transfer question

Deleting vertices 8, 9, 19, 20 leaves a twenty-vertex graph that does admit two ordinary gear contractions. Exact reconstruction identifies a nine-vertex edge-subdivided seed of stable bound three, a fourteen-vertex intermediate of bound four, and the twenty-vertex core of bound six. The first row transition is proper g-lifting; the second is proper geared lifting with the left hubs doubled, in the terminology and formulas of [10, Definitions 2.4 and 2.7]. Reintroducing the four deleted vertices in the displayed order gives maximal classical lifting coefficient one each: every restricted stable optimum is five. Therefore neither the classical row nor its construction is claimed new. These classical operations do not themselves prove that squared quantum expectations satisfy the lifted rows. Theorem 7.2 settles one final instance, without establishing a quantum transfer rule.

A preregistered small screen tested proper geared unit-seed rank rows on 30 selected constructions from connected graphs of orders three through six. There were 240 starts, capped at 32 steps each; none met the strict convergence test. No violation above 10^{-6} was found. Independent bitwise checks covered 31 saved best states including a positive control of $3.0445923942 > 3$. This is weak falsification evidence, not a global optimization result. It omits nonuniform seed facets, proper g-lifting, and the four XX vertex additions.

8 Verification, scope, and publication boundary

The manuscript draws on research snapshot 4dc9dc7; the full commit is in Appendix B, with later results through C034 added separately. Discovery and acceptance are separate programs. Universal algebra is checked by exact word rewriting and integer amplitude polynomials. Polyhedral discovery uses rational double description; independent acceptance clips every halfspace from the full rational unit cube and verifies the final vertices and facet ranks. Missing-facet and corrupted-certificate controls must be rejected. Numerical states are checked independently through their bitwise Pauli action.

The accepted clean-source reproduction used code/data commit a7a92f3, followed by documentation-only commit 4dc9dc7. All 127 tests passed in one run in 441.989 seconds, and all 64 artifact hashes matched. C011, C012, C014 and C016 discovery outputs were regenerated; saved C015 states, not every optimization trajectory, were reevaluated. These are same-host checks with different algorithms, not external replication or proof-assistant verification.

For C031 separately, a clean scoped archive of commit 08b71d4 passed the standalone standard-library exact verifier and eight C031 tests. The first archive attempt exposed a CRLF/LF source-hash mismatch; explicit LF normalization fixed portability without changing any coefficient. This later check did not rerun the historical 127-test campaign or all discovery trajectories. Later C033 construction tests and C034 state-audit tests are recorded separately.

Failed attempts remain in the ledger. A complete hull enumeration for Y exceeded a 300-second cap and is not accepted as complete. The full-facet claim in Proposition 7.1 instead uses exhaustive stable-set enumeration for just the frozen weight. For X^+ , two clipping attempts timed out before acceptance; an exact candidate-edge index eventually passed both the full hull and missing-facet checks. The index retains the original exact adjacency rank test and falls back to full pair scanning when its index budget is exceeded.

The established beta framework, closure rules and state-polynomial relaxations are prior art [7]. XX-strips are published classical objects [9]; gear composition describes classical stable-set polytopes under specific hypotheses [10]. These facts must not be relabeled as new quantum

mechanisms. A quantum all-weight result for the specified family is the candidate mathematical contribution, but equivalence to a consequence of a larger known closure or embedding remains a priority question. We do not infer novelty from a failed keyword search. The wider source comparison and unresolved applicability questions are retained in the repository’s novelty audit. The line-graph theorem is distinct: it covers a general free-fermion class and gives a maximum-weight-matching oracle, but it does not imply the non-line SCF results below.

The quantum consequences proved here are tight uncertainty bounds: for every proved graph,

$$\sum_i w_i \text{Var}_\rho(P_i) \geq \sum_i w_i - \alpha(G, w),$$

since $P_i^2 = I$. They also give valid Hamiltonian norm bounds by (2) and weighted Cauchy–Schwarz. These are standard consequences of all-weight equality, not demonstrated runtime or sample-complexity advantages. No paid cloud experiment or QPU job was performed. The certified composition facet is neither a hardware advantage claim nor a completed universal SCF theorem.

9 Conclusion

Nuclear-norm duality and Edmonds’ matching polytope prove that every line graph is $\hbar$ -perfect for all nonnegative weights. The result contains an explicit infinite family outside the h -perfect sufficient class and turns its weighted Pauli uncertainty problem into maximum-weight matching. A coherent signed Gram representation and an exact fixed-core refinement prove all-weight Pauli uncertainty for an explicit unbounded SCF family. The same scalar envelope completes a three-row atom and supplies two XX-based corollaries. The composed twenty-four-operator target now has an exact rational quantum upper certificate matching its classical bound six, despite a certified first-moment relaxation gap. Its classical gear/lifting ancestry is established. The remaining boundary for the broader SCF program is a conceptual quantum composition theorem. External review and a broader priority assessment remain necessary before a first-discovery or venue-level claim.

A Explicit graph and operator conventions

The XX graph X uses the cycle 1–2–3–4–5–6–1 and the following additional adjacency lists, each adding only the listed edges:

7	1, 2
8	4, 5
9	6, 1, 2, 3
10	3, 4, 5, 6, 9
11	3, 4, 6, 1, 9, 10
12	2, 3, 5, 6, 9, 10
13	1, 2, 4, 5, 7, 8.

It has 37 edges. The graph6 strings of X and X^+ , respectively, are

LhEM?rcNLhleuo NhEM?rcNLhleuo?_?GG.

The two-core graph Y has graph6

WhENH}vZuo???@???b@GoP?@C?FG?Bw?J[?B]EBkE?O???oB.

The authoritative Pauli labels for Y are integer pairs (x, z) in `scf_two_xx_weight_c014.json`. For a seven-bit basis label j ,

$$P(x, z)|j\rangle = i^{\text{popcount}(x \& z)} (-1)^{\text{popcount}(j \& z)} |j \oplus x\rangle.$$

This formula fixes the Hermitian phase and bit ordering. Pairwise anticommutation is the binary symplectic product $x \cdot z' + z \cdot x' = 1$ over $\mathbb{F}_2$. Every such product is checked. Stored complex states encode each amplitude as a pair of real and imaginary floating-point components. Their acceptance is numerical with an explicit tolerance, unlike rational hull and Gram certificates.

B Certificate inventory and reproduction

The public repository is <https://github.com/YaroslavPelekhov/Quantum>. The original research snapshot used for this draft is

4dc9dc7bc26cca5ea3a7f4cfa66cc4f7fc969a83.

Source checkers are under `experiments/pauli_fourth_moment_phase0/`; data are under `results/pauli_fourth_moment_phase0/`. The latter directory’s `manifest.json` lists the 64 earlier artifacts’ canonical SHA-256, not the later C031–C034 additions. The exact-six archive audit used 08b71d4; subsequent research through C034 is in efe5ea0. JSON canonicalization here only normalizes CRLF to LF; it does not reorder keys or recompute numerical values.

Obligation	Principal artifacts / acceptance programs
Uniform Gram	<code>scf_rectangular_gram_bridge.json</code> ; <code>verify_scf_rectangular_gram_bridge.py</code>
Uniform hull/core	<code>scf_uniform_facet_gate.json</code> , <code>scf_core_refinement.json</code> ; <code>verify_scf_uniform_facet_gate.py</code> , <code>verify_scf_core_refinement.py</code>
Three-row proof	<code>scf_three_row_gram_c009.json</code> ; <code>verify_scf_three_row_gram.py</code> (also checks the target hull)
XX atom	<code>scf_xx_gate_c011.json</code> ; <code>verify_scf_xx_gate.py</code>
Closed XX	<code>scf_xx_closure_c012.json</code> ; <code>verify_scf_xx_closure.py</code>
Joint facet	<code>scf_two_xx_weight_c014.json</code> ; <code>verify_scf_two_xx_weight.py</code>
Physical attack	<code>scf_two_xx_attack_c015.json</code> ; <code>verify_scf_two_xx_attack.py</code>
Theta witness	<code>scf_two_xx_baseline_c016.json</code> ; <code>verify_scf_two_xx_baseline.py</code>
Exact quantum six	<code>c031_exact_six_certificate.json</code> ; <code>verify_c031_exact_six.py</code> , <code>test_c031_exact_six.py</code>
Classical ancestry	<code>c033_gear_bridge_with_lifting.json</code> ; <code>run_c033_gear_bridge.py</code>
Transfer screen	<code>c034_gear_falsification.json</code> ; <code>verify_c034_states.py</code>
Line graphs	<code>c038_line_graph_hbar.json</code> ; <code>verify_c038_line_graph_hbar.py</code> , <code>test_c038_line_graph_hbar.py</code>
Integrity	<code>manifest.json</code> ; <code>verify_scf_research_bundle.py</code>

For example, from the repository root run the following commands (replace the short filename in the second line for the other exact checkers listed above):

```
python -S experiments/pauli_fourth_moment_phase0/
    verify_scf_research_bundle.py
python -S experiments/pauli_fourth_moment_phase0/
    verify_scf_three_row_gram.py
```

The displayed wrapped paths must be joined into one line each. All independent exact acceptance programs use the standard library. The full regression suite additionally imports NetworkX and must *not* use `-S`:


```
python -m unittest discover
-s experiments/pauli_fourth_moment_phase0
-p "test_scf*.py"
```

Assertions are part of certificate acceptance: never invoke these checkers with `-O`. Discovery additionally uses the recorded NumPy/SciPy/SymPy/CVXPY environment and `cdd.gmp` for exact hulls. Rediscovery should run in a new archive rather than overwrite reference artifacts. The registered C015 attack is not extended by this manuscript.

The paper-specific checker compares the headline constants with the archived JSON and verifies the claim-status flags. It supplements, not replaces, the mathematical checkers. Manuscript authorship, affiliations, funding, and final submission disclosures must be approved by the human authors before submission. This is an AI-assisted working draft, not a submitted or externally reviewed article.

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